



GE Medical Systems

Technical Publications

Direction 2209613

Revision 1

Signa[®] Horizon_t HiSpeed with SGD to EchoSpeed with SGD Upgrade

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Operating Documentation

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Call Traffic and Transportation, Milwaukee, WI (414) 785-5052/8*323-5052 **immediately** after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section "S" of the Policy & Procedure Bulletins.

3/12/92



GE Medical Systems

**GE Medical Systems:
Telex 3797371
P.O. Box 414, Mil-
waukee, Wisconsin**

REVISION HISTORY

<u>REV</u>	<u>DATE</u>	<u>PRIMARY REASON FOR CHANGE</u>
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Rev 0	.. Mar 11, 1998	... Initial Release
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Rev 1	.. Jul 19, 1999	... Added Gradinet Configuration Change in Section 2
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SECTION 1 – OVERVIEW

1-1 SIGNA HORIZON SYSTEM CONFIGURATION

This manual provides instructions to upgrade a Signa Horizon HiSpeed with SGD System to Signa Horizon EchoSpeed with SGD System. To install this upgrade on a Horizon 5.x system, Release 5.7 (or later) software is required. This installation manual is applicable to Fixed-Sites at magnet field strengths of 1.5T.

This Direction covers the following catalogs:

- M1090WG – Horizon HiSpeed SGD to EchoSpeed SGD Upgrade

To complete the installation of this upgrade, one of the two following Receiver 0 catalogs is required:

- M1090WJ – Receiver 0 for 8.x and 8.x w/SGD Sites without Spectro that are upgrading to EchoSpeed
- M1090ZL – Receiver 0 for 5.x and 5.x w/SGD Sites without Spectro that are upgrading to EchoSpeed

Note

There are some naming differences between the sales literature documents and the technical documentation the catalog in this section. Listed below are the corresponding names. The Pre-Installation document uses the technical document names.

SALES LITERATURE NAME = TECHNICAL DOCUMENT NAME

SX Gradient Subsystem	=	SGD (Scaleable Gradient Driver)
Signa Horizon	=	Signa Horizon with SGD or Signa Horizon 5.x with SGD
Signa Horizon LX	=	Signa Horizon LX with SGD or Signa Horizon 8.x with SGD

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion. Refer to *Direction 2142460, Signa Horizon Upgrade Pre-installation*, Pre-installation Checklist tab.

Tools and test equipment to install and calibrate the Signa Horizon system are listed in *Direction 2142460, Signa Horizon Upgrade Pre-installation*, Section 10, TOOLS AND TEST EQUIPMENT.

1-3 PRODUCT LOCATOR

Product Locator System tracks creation, shipment, trans-shipment, and installation location of serialized models.

The installation card is not a “warranty card”, therefore it is to be processed as soon as the installation starts.

The PLC system also includes Remanufacture, Scrap, and Transfer cards. They should be obtained and completed for all serialized cabinets removed from the system according to Policy and Procedure.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. Process just the installation card and discard any extra shipment cards and labels.

System upgrade serial-number rating plates are furnished. They provide the means to track systems with various hardware versions of the System Upgrade. Kit added rating plates are furnished with various upgrade kits. They signify that model numbers on the major component have changed function and compatibility and have become a “look-alike” for current production in performance and compatibility. These rating plates do not have shipment cards since serial numbers are not assigned.

1-4 DISCARDED MATERIAL RETURN POLICY

The “discard” or “return pile” consists of items removed by the Signa Horizon Upgrade or traded in for new options and have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the return pile is per GE Medical Systems Policy and Procedures.

1. For coordination of return pile, contact GEMS Recycling Center: Phone (414)–747–6997 or Dial Comm 8*579–6997; Fax (414–747–6855); WizMail “RECYCLING”; Field ASPEN 1–800–525–1516, Box #26041.

When calling, the following information is needed:

- FDO (Future Delivery Order) number under which the upgrade was shipped
 - Site name, address, and phone number
 - Date of completed Horizon upgrade.
 - Name and Beeper Number of Field Engineer that could be available when the equipment is picked up.
2. Prepare the return pile for shipment as follows:
 3. Process product locator cards for all serialized components removed from the system according to policy.
 4. Regulations governing the disposal of this material in the recycle pile vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items in the “recycle pile” per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1-5 UPGRADE SEQUENCE

Follow the Horizon system Upgrade Flow chart in SECTION 2 for best installation efficiency.

Note that many procedures may be performed in parallel and may be performed in any order according to the specific situation of each site. However all parallel tasks are to be completed prior to the next task.

The general sequence of events is:

- Verify that the upgrade has arrived complete and site is ready to shut down.
- Shut System cabinet down.
- Install EchoSpeed upgrade
- All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy
- Power up
- Functional Checks
- System calibrations
- System performance checks
- Cover replacement
- Complete final tasks and return site to customer.

1-6 SYSTEM CABINET POWER SHUTDOWN

ELECTRIC SHOCK HAZARD!! ENSURE THAT POWER IS OFF TO THE INVOLVED UNITS BEFORE STARTING THIS PROCEDURE.



Static electricity can damage circuit boards and other delicate electrical equipment. Always use appropriate ESD procedures and an ESD Kit. Use a grounding wrist strap and ESD Storage Bag.

1. Shutdown Signa software.
2. Turn Power OFF Power at the System Cabinet.
3. In the PDU, turn OFF circuit breaker CB11, System Cabinet.
4. Lock and tag PDU Front Door.
5. Verify Power is OFF at the System Cabinet.



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1-7 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Record and enter applicable data into applicable site configuration files and records.
3. Process product locator installation cards for all serialized components added to the system to include product locator card supplied with the upgrade serial number rating plate. Return to:

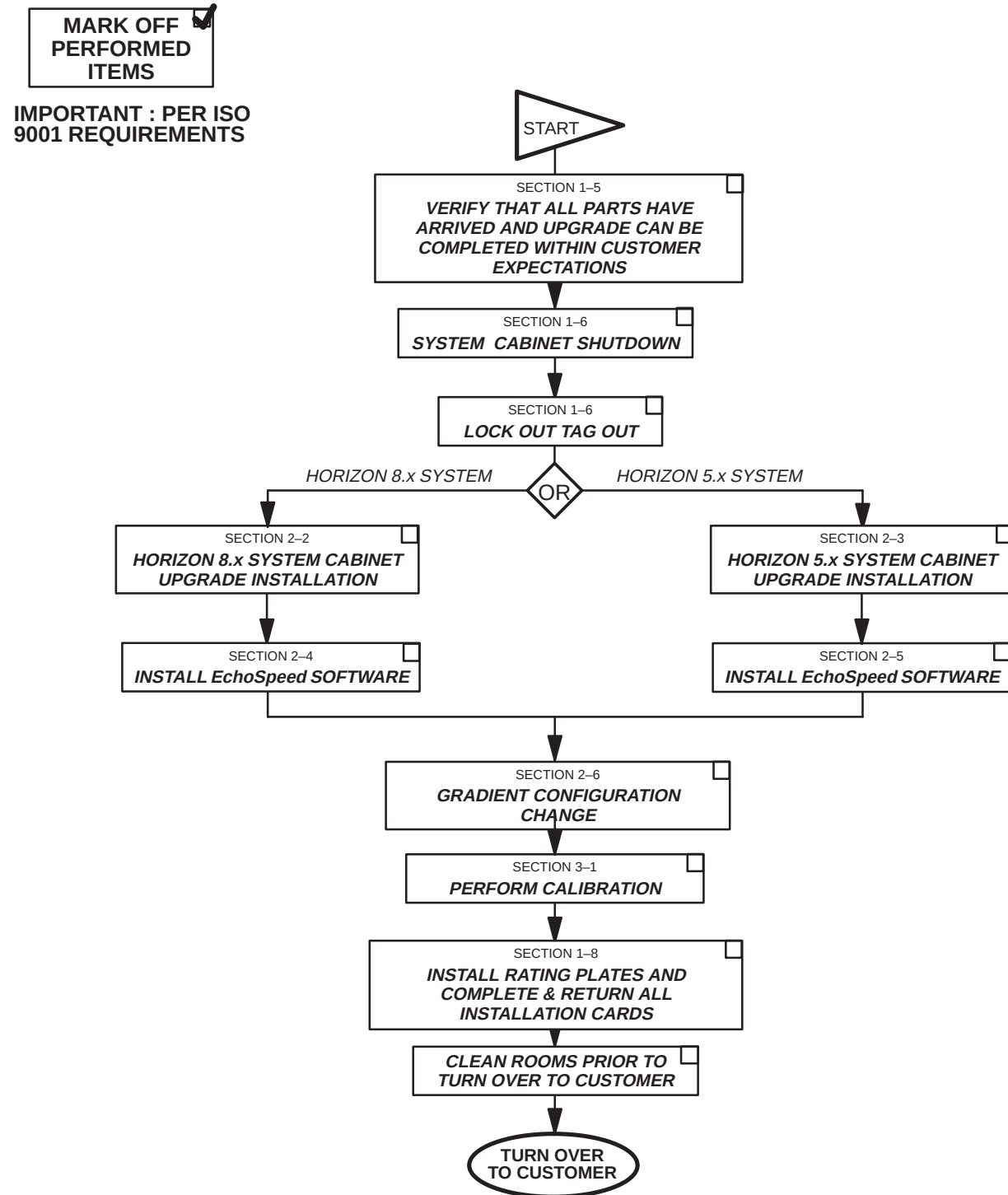
**Product Locator File
P.O. Box 414, W-523
Milwaukee, WI 53201-0414**

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

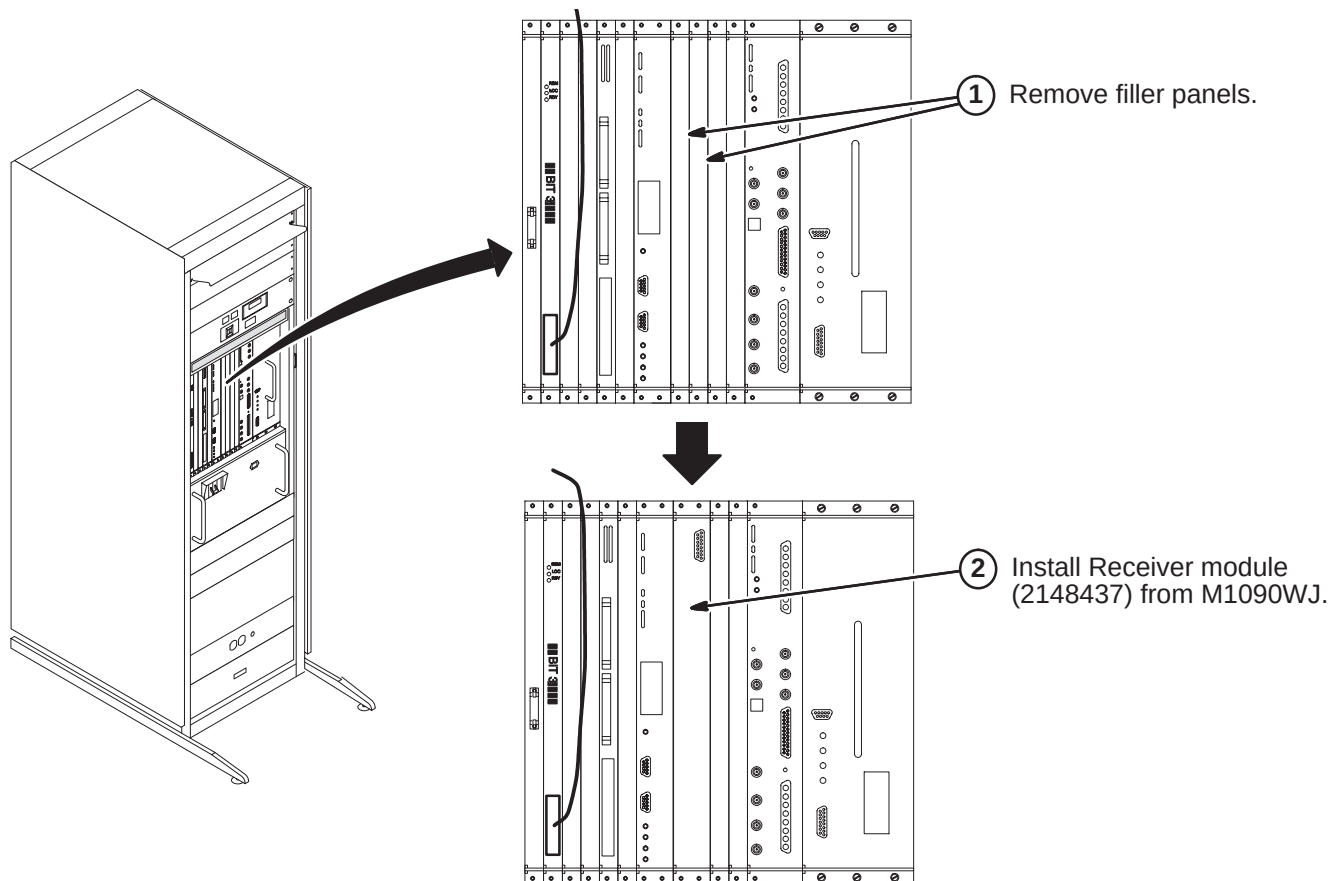
SECTION 2 – MECHANICAL UPGRADE PROCESS

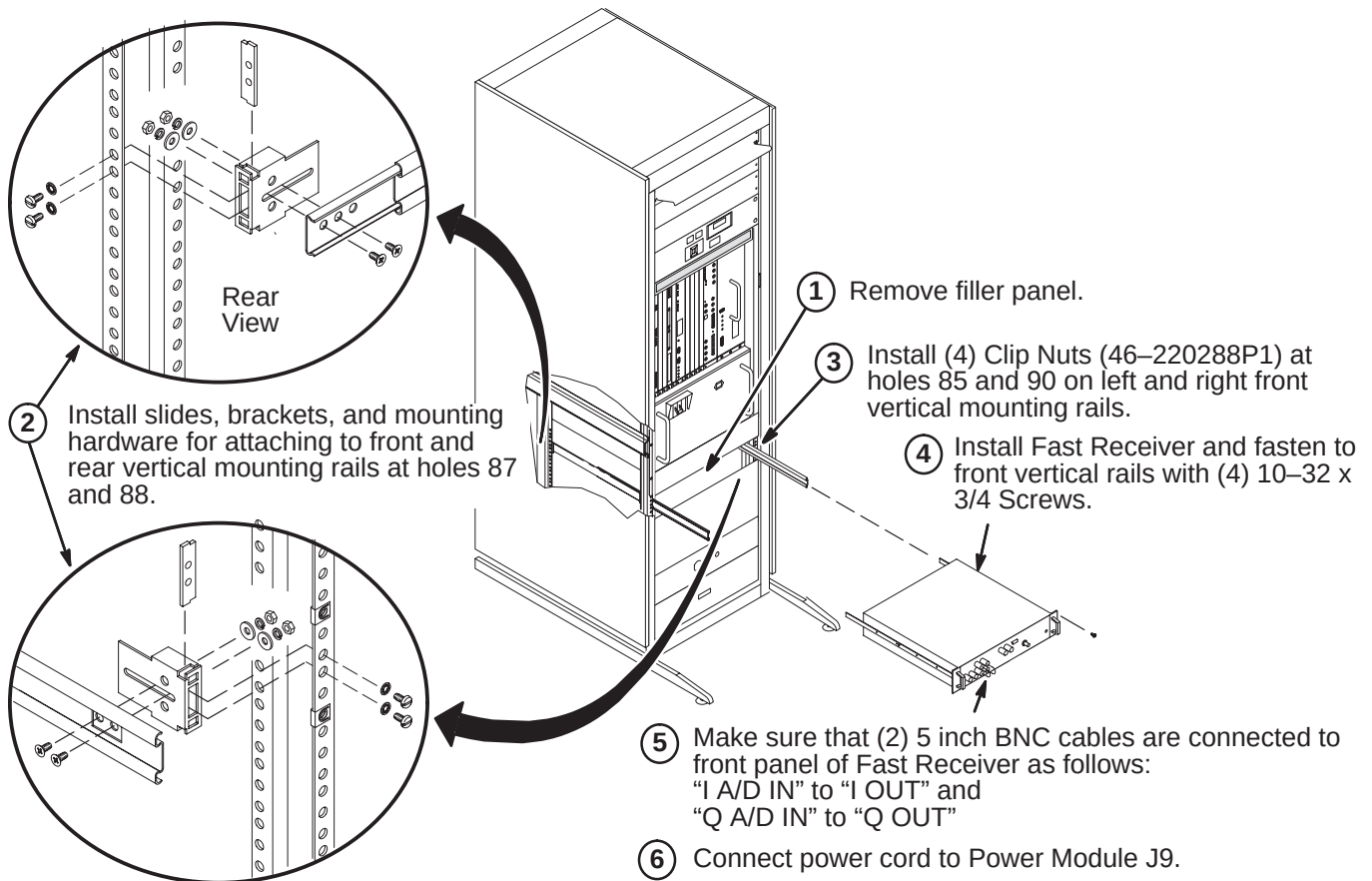
2-1 INSTALLATION FLOWCHART



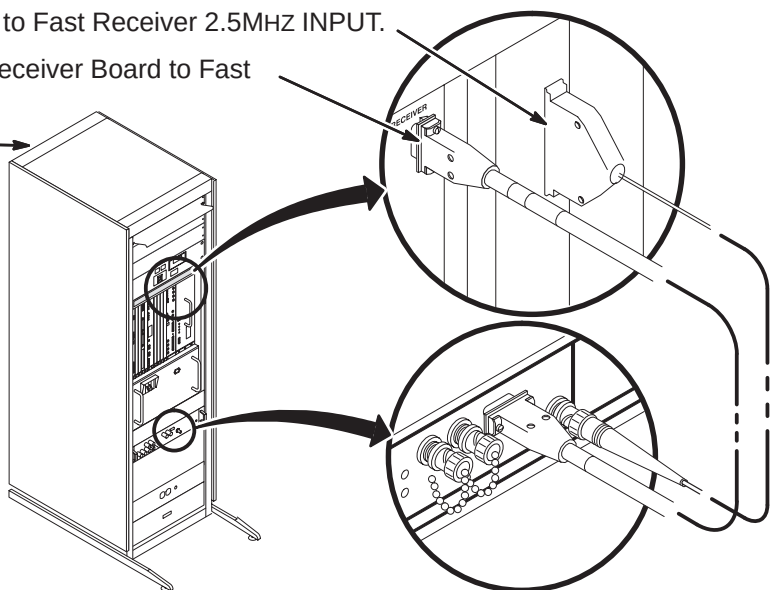
2-2 HORIZON 8.x SYSTEM CABINET UPGRADE INSTALLATION

2-2-1 Install Receiver Module



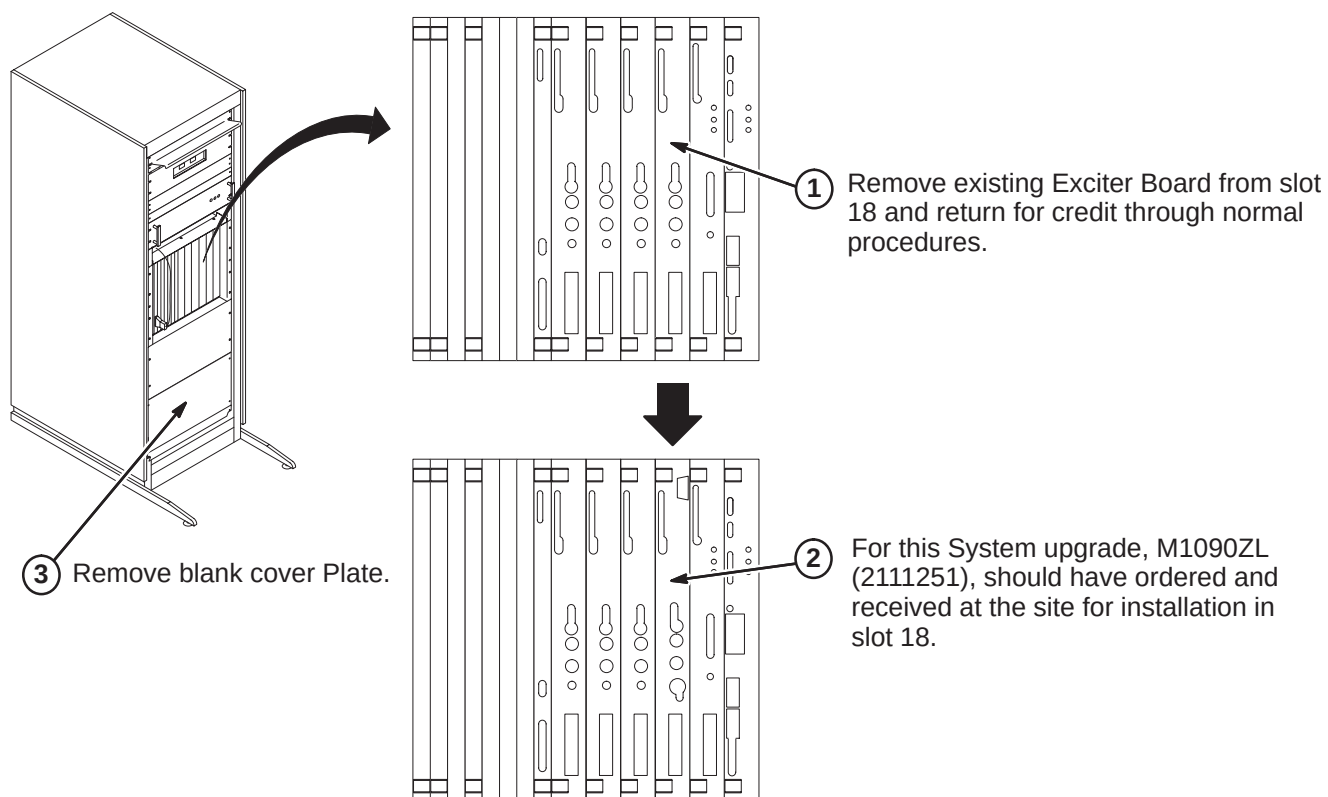
2-2-2 Install Fast Receiver Module**2-2-3 Install Fast Receiver Cables and Rating Plate**

- 1** Connect Cable (2144527) from CERD J101 to Fast Receiver 2.5MHZ INPUT.
- 2** Connect Digital Interface Cable from Fast Receiver Board to Fast Receiver DIGITAL INTERFACE.
- 3** Apply Rating Plate (46-302200P12) above rear door,
- 4** Notify field service and other installation personnel that are working at the site that locks and tags will be removed so power can be turned on.
- 5** Restore power to the System cabinet.



2-3 HORIZON 5.x SYSTEM CABINET UPGRADE INSTALLATION

2-3-1 Install Fast Receiver



2-3-2 Install Fast Receiver Module

- ① Install slides, brackets, and mounting hardware for attaching to front and rear vertical mounting rails at holes 87 and 88.
 - ② Install (4) Clip Nuts (46-220288P1) at holes 85 and 90 on left and right front vertical mounting rails.
 - ③ Attach slide with provided mounting hardware and install Fast Receiver.
 - ④ Install Fast Receiver and fasten to front vertical rails with (4) 10-32 x 3/4 Screws.
 - ⑤ Make sure that (2) 5 inch BNC cables are connected to front panel of Fast Receiver as follows:
"I A/D IN" to "I OUT" and
"Q A/D IN" to "Q OUT"
 - ⑥ Connect power cord to Power Module J9.
 - ⑦ Install (4) Clip Nuts (46-220288P1) at holes 92 and 101 on left and right front vertical mounting rails. Attach blank Cover Plate (46-282213P1) to Nut Clips using (4) 10-32 x 3/4 Screws (46-170015P62).
-

2-3-3 Install Fast Receiver Cables and Rating Plate

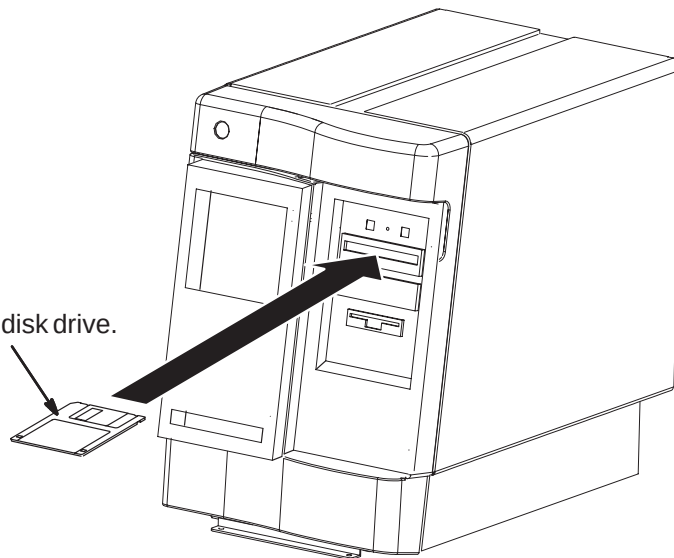
- ① Connect Digital Interface Cable (2137306) to Receiver-0 J14.
 - ② Connect BNC Cable (2113888) to Receiver-0 SMB J11.
 - ③ Apply Rating Plate (46-302200P12) above rear door,
 - ④ Notify field service and other installation personnel that are working at the site that locks and tags will be removed so power can be turned on.
 - ⑤ Restore power to the System cabinet.
-

2-4 SIGNA HORIZON LX SOFTWARE OPTIONS INSTALLATION (Using MOD Drive)

The EchoSpeed Option (2210001) resides on a MOD disk.

① Insert MOD disk (2209991) into the host (SGI) MOD disk drive.

② Refer to Direction 2209382, supplied with MOD disk, for instructions on installing EchoSpeed software option.



2-5 SIGNA HORIZON 5.x EchoSpeed SOFTWARE INSTALLATION

The MOD disk is not required for Signa Horizon 5.x systems and can be discarded. The EchoSpeed option is enabled when the Release 5.7 (or later) software is loaded.

2-6 GRADIENT CONFIGURATION CHANGE

The default Gradient Configuration File needs to be changed to reflect the slew rate parameters for SR120.

Open a C shell. For Release 5.x Systems, Touch **UTILITIES**, then **C Shell**. For Release 8.x Systems, Select the Service Desktop, then select C Shell.

In the C Shell type:

cd/w/config <Enter> (set directory)

mv GradientConfig.cfg GradientConfig.cfg.old <Enter> (renames default file)

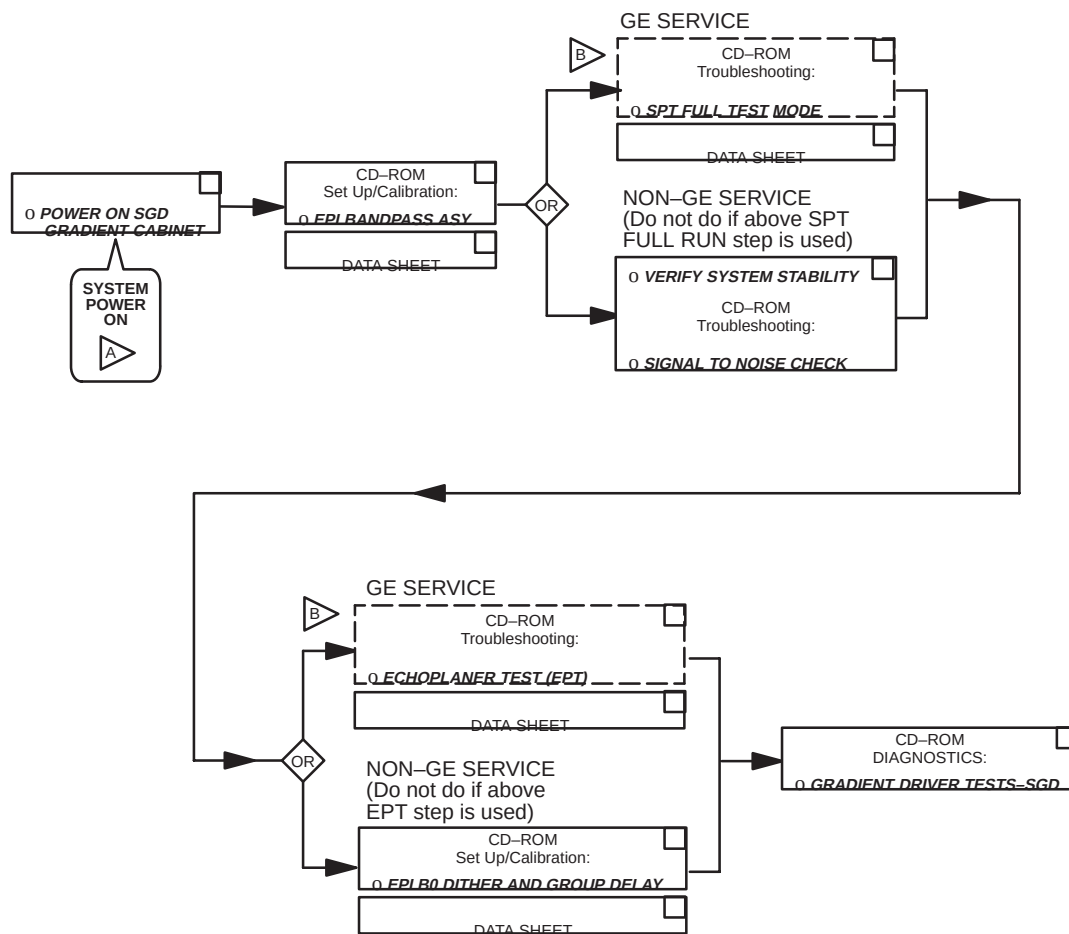
cp GradientConfig.cfg.120.ECHOSPEED_SGD GradientConfig.cfg <Enter> (copies the SR120_SGD file as default)

ls -al Gradient* (to verify the changes just made)

exit <Enter> (to close the C Shell)

SECTION 3 – CALIBRATION PROCESS

3-1 CALIBRATION FLOWCHART



A DO NOT TURN ON THE **SGD HI-SLEW** CABINET CIRCUIT BREAKERS (BREAKERS ON FRONT BOTTOM OF CABINET) UNTIL ROOM HUMIDITY IS AT 60% OR LESS.

CAUTION! EQUIPMENT IS SUSCEPTABLE TO FAILURE IN HIGH HUMIDITY.

B INDICATES AN ADDITIONAL PROPRIETARY TEST OR ALTERNATIVE PROPRIETARY CALIBRATION PROCEDURE AVAILABLE FOR GE USE OR TO CUSTOMERS WITH AN ADVANCED SERVICE PACKAGE LIMITED LICENSE.